

Refresh & Content Opportunity Scoring: A Leak-Free Offline Action Engine on Search Telemetry

Capstone Research Paper · FlyRank Applied Machine Learning Internship (ML-CAP-01)

Author: Muhammad Arsalan	Track: Machine Learning Capstone (Week 8)	Target Metric: Precision@50 (Client-Holdout)
Dataset: 30,000 Scored Enterprise URLs	Deployed Paper: arslanflyrankweb1.netlify.app/paper.html	Data Credit: flyrank.ai

ABSTRACT: In enterprise search marketing, organizations manage tens of thousands of published URLs with capacity to refresh only 20 to 50 articles weekly, making unranked or recency-sorted audits unscalable. We investigate whether pre-decision search performance signals—specifically impression continuity, striking-distance position tiers, and content staleness—can reliably rank candidate URLs for organic traffic recovery on completely unseen client domains. Utilizing an anonymized dataset of 30,000 URLs across multiple commercial verticals, we enforce a strict pre-observation data contract and evaluate via GroupShuffleSplit on client_id to eliminate target leakage and domain memorization. Our tuned Random Forest achieves **Precision@50 = 0.740** (ROC-AUC = 0.750, PR-AUC = 0.618), delivering a **3.08× lift over the transparent heuristic baseline (0.240)** and converting triage queues into high-confidence editorial feeds. The resulting action engine outputs ranked recommendations accompanied by human-readable diagnostic reason codes, mapping URLs to refresh, CTR optimization, or monitoring playbooks without asserting causal ranking claims.

1. Introduction & Problem Statement

Digital content assets degrade over time as search query semantics drift, competitor content expands, and ranking algorithms recalibrate. However, identifying which declining articles warrant editorial intervention remains a critical bottleneck. Enterprise editorial teams operate under finite capacity—typically able to refresh 20 to 50 URLs weekly. Sorting URLs by age wastes effort on dead content with zero search demand, while sorting by raw traffic wastes time reviewing healthy evergreen pages. This research establishes a leak-free offline machine learning engine that scores and ranks candidate pages for refresh review, maximizing the proportion of genuine decay opportunities identified within the weekly top-50 triage window.

2. Data & Public-Safe Sanitization

The analysis is conducted on the anonymized FlyRank search dataset (data/raw/content_refresh_anonymized.csv), comprising 30,000 URLs across five commercial client domains over a 90-day pre-decision historical aggregation window. The dataset contains 16,262 declining URLs (is_declining_label = 1), establishing an overall positive base rate of 54.21%. In compliance with public research rules, zero client names, domains, URLs, or private search queries appear in features or models.

3. Methodology & Leak-Free Validation Design

The Leakage Trap: Initial naive features (trend_pct) achieved artificial 1.000 Precision@50 because they were calculated across the outcome window. We enforced a strict pre-observation data contract restricting signals to: days_with_impressions (demand continuity), log_impressions_90d (volume scale), avg_position (SERP rank), content_age_days (staleness), and scroll_rate (engagement).

Grouped Client Holdout: Standard random splits leak site-wide taxonomy. We enforce GroupShuffleSplit grouped strictly on client_id (80% train, 20% test). Unseen client domains exist exclusively in test.

Heuristic Baseline: Flagging stale content (content_age_days > 180) situated in striking-distance rankings (positions 4–20).

4. Results & Model-vs-Baseline Benchmarking

Model Specification	Precision@50	ROC-AUC	Avg Precision	Recall	F1 Score
Random Forest (n=100, d=10)	0.740 (3.08× Lift)	0.750	0.618	0.744	0.640
Decision Tree (max_depth=5)	0.540	0.742	0.575	0.716	0.634
Logistic Regression (L2, scaled)	0.400	0.700	0.522	0.567	0.566
Week-4 Heuristic Baseline	0.240	0.627	0.468	—	—
Dataset Test Base Rate	0.534	0.500	0.534	1.000	0.696

Key Finding: Random Forest delivers **Precision@50 = 0.740**, representing a **3.08× lift over baseline (0.240)**. 37 out of 50 recommended URLs represent genuine decay opportunities, eliminating over 70% of wasted editorial reviews.

Feature Hierarchy: days_with_impressions (15.78%), log_impressions_90d (12.82%), avg_position (10.90%), content_age_days (9.55%), and char_count (4.26%).

5. Limitations & Honest Scientific Framing

- **Non-Causal Decision Support:** The model identifies correlation with observed traffic decline; it does not prove causal ranking recovery.
- **Domain Generalizability:** Validated across commercial domains; fine-tuning required for technical docs or news aggregators.
- **Error Analysis:** False positives occur on high-authority articles (age > 400d, pos 8–12) buffered by strong backlink equity. False negatives stem from sudden seasonal query drift on fresh articles (age < 90d) lacking long-term staleness signals.

6. Ranked Recommendations (Action Playbooks)

Action Playbook	Queue Count	Diagnostic Reason Code	Editorial Intervention Protocol
refresh	8,178 (27.3%)	declining_with_demand	Complete factual overhaul; update dates, references, statistics.
refresh_and_review_ctr	6,657 (22.2%)	low_ctr_visible_page	Title tag and meta description rewrite to capture SERP clicks.
refresh_and_review_engagement	1,990 (6.6%)	low_engagement_visible_page	Restructure above-the-fold readability and reduce bounce rate.
expand_and_refresh	82 (0.3%)	high_demand_striking_distance	Substantial content expansion targeting page-1 rank breakthrough.
monitor	13,093 (43.6%)	stable_or_recovering	No action required. Preserve existing evergreen ranking equity.

7. Reproducibility & Acknowledgments

Reproducibility: All notebooks and pipelines are version-controlled at github.com/24pwai0015-max/flyrank-ml-muhammad-arsalan.

Data Credit: Built on the *FlyRank ML Internship dataset*. We thank **FlyRank.ai** (<https://flyrank.ai>) for supplying enterprise search telemetry and computing infrastructure.